

The Impact of Weather Extremes on Firm-Level Productivity

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Abstract

Climate change is intensifying weather extremes worldwide, yet our understanding of how these shocks affect firm-level productivity remains limited. This paper provides the first comprehensive assessment of how diverse weather dimensions impact economic productivity using granular firm data across 50+ countries. We match over 20 million firm-year observations with high-resolution weather data, examining the economic implications of a broad range of weather statistics, including temperature variability, heatwaves, and cold spells. We find that, beyond the well-studied inverted U-shaped relationship between mean temperatures and productivity, heatwave severity is the most economically significant weather determinant: a one-standard-deviation increase reduces firm turnover per employee by approximately 0.25%. Coldwaves also significantly reduce productivity, by 0.19% per standard deviation. Day-to-day temperature variability, while negatively associated with productivity, is not statistically significant in the full sample. Decomposing results by Köppen climate zones reveals important heterogeneity: heatwaves are harmful across all climate zones, temperature variability is significant only in temperate climates, and coldwaves are concentrated in cold and arid zones. Supplementary adaptation analysis, reported in the Appendix, suggests partial adaptation to temperature variability and precipitation but no adaptation to extreme heat. These results provide crucial microeconomic foundations for climate damage functions and suggest that focusing exclusively on mean temperature changes underestimates the economic costs of climate change, particularly as extreme heat events become more frequent and intense.

Keywords: Climate change damages; Extreme weather; Firm performance; Weather shocks.

JEL Classification: Q54; C23; E24.

1 Introduction

Climate change poses fundamental challenges to economic prosperity and stability worldwide. Rising global temperatures and shifting precipitation patterns are increasingly disrupting production processes, supply chains, and labor markets. While the aggregate effects of climate change on economic growth have been documented in macroeconomic data ([Burke et al., 2015](#); [Kalkuhl and Wenz, 2020](#); [Kotz et al., 2021](#); [Kahn et al., 2021](#)), the specific mechanisms through which weather extremes affect firm-level productivity remain incompletely understood. Given that productivity is the main driver of sustained economic growth and rising incomes over the long run ([Solow, 1956](#)), understanding how climate extremes impact firm performance has important implications for both climate policy and economic planning.

Weather extremes, including cold waves, heat waves, floods, droughts, and storms, are particularly harmful to the economy as they create sudden disruptions in production processes, damage infrastructure, and alter both supply and demand conditions. Unlike gradual temperature increases, extreme weather events impose immediate shocks on firms through multiple channels: worker absenteeism due to transport disruptions, reduced cognitive and physical capacity of workers, damage to equipment and inventory, and interruptions to energy and water supply. These events are projected to increase substantially in both intensity and frequency under climate change scenarios, even those consistent with ambitious mitigation targets (IPCC, 2023). Despite this growing threat to economic activity and livelihoods, large-scale empirical evidence on the impacts of diverse weather extremes across different climates, sectors, and economic contexts remains limited. Most existing firm-level studies have focused on specific countries or regions ([Somanathan et al., 2021](#); [Zhang et al., 2018](#)), examined single types of weather extremes (typically heat stress, e.g. [Costa et al. \(2024\)](#)), or relied on aggregate macroeconomic data that obscures important heterogeneities in firm-level responses ([Akyapi et al., 2024](#)).

A distinct but related question concerns adaptation. The distinction between weather and climate is conceptually important: climate represents the statistical pattern of weather that persists over time and to which agents can adapt, while weather realizations represent unpredictable deviations from this pattern ([Traeger, 2025](#)). This framework suggests that welfare-relevant damages arise primarily from weather shocks rather than from predictable climate patterns to which agents have already adjusted. In health outcomes, adaptation appears effective: [Carleton et al. \(2022\)](#) show mortality-temperature relationships flatten with income and local climate adaptation, and [Barreca et al. \(2016\)](#) document declining US heat mortality as air conditioning spread. For labor markets, [Robinson et al. \(2024\)](#) find heat alerts

reduce productivity losses, though only partially. Yet systematic evidence is less encouraging: [Burke et al. \(2024\)](#) find that only six of 21 outcomes show declining climate sensitivity over half a century, while five show *increasing* sensitivity. [Callahan and Mankin \(2022\)](#) document that extreme heat damages remain globally unequal, reaching 8% of GDP annually for the poorest regions versus 3.5% for the wealthiest. Firm-level evidence on adaptation—particularly to extreme heat—remains largely absent.

This paper addresses this knowledge gap by estimating the responsiveness of firm performance to a comprehensive set of weather extremes using granular, firm-level data across diverse climates and economic contexts. Our primary measure of firm performance is turnover per employee (labor productivity), which captures the efficiency with which firms generate revenue relative to their workforce and is directly comparable across countries, sectors, and firm sizes. We examine how this measure responds to various dimensions of weather extremes, allowing us to identify which types of climate shocks pose the greatest threats to economic productivity.

To conduct this analysis, we assemble a novel dataset combining detailed firm-level financial information with high-resolution weather data. Our firm-level data come from the ORBIS database, which provides comprehensive balance sheet information for millions of firms across Europe and beyond. This database covers firms of all sizes - from micro-enterprises to large corporations - and spans manufacturing and services sectors, offering a large and heterogeneous pool of firms that operate under diverse climatic conditions and economic environments. We match these firms to their precise geographic locations and construct measures of weather extremes at each location using the ERA5-Land reanalysis product from the European Centre for Medium-Range Weather Forecasts. We build several indicators capturing different dimensions of weather extremes that theory and prior case studies suggest should affect productivity: cold waves (periods of sustained below-normal temperatures), total precipitation, precipitation extremes (heavy rainfall events), the number of wet days, potential evapotranspiration (a measure of atmospheric water demand and drought stress), snowfall, and daily temperature variability. We also include more canonical measures of annual mean temperatures and maximum temperatures to capture gradual climate shifts. This comprehensive approach allows us to move beyond the focus on heat extremes that has characterized much of the literature and to assess the relative importance of different types of weather shocks for firm productivity.

Our findings make several important contributions to understanding the economic consequences of climate change. First, we provide the most comprehensive firm-level assessment to date of how di-

verse weather dimensions, not just temperature, affect economic productivity across multiple countries and climatic zones. We identify heatwaves and coldwaves as the dominant weather determinants of firm productivity: a one-standard-deviation increase in heatwave severity reduces turnover per employee by 0.25%, while coldwaves reduce it by 0.19%. These findings confirm and extend the results of [Costa et al. \(2024\)](#) on the detrimental role of heatwaves, while revealing that cold extremes—previously overlooked—also impose significant economic costs. Day-to-day temperature variability, highlighted in macro-level studies by [Kotz et al. \(2021\)](#) and [Linsenmeier \(2023\)](#), is negatively associated with productivity but not statistically significant in the full sample, suggesting that its effects may be more localized.

Second, we decompose the analysis by Köppen climate zones, revealing important heterogeneity in weather impacts. Temperature variability significantly reduces productivity only in temperate climates, while coldwaves are concentrated in cold and arid zones. Heatwaves, by contrast, are harmful across all climate zones, underscoring the universal economic threat posed by extreme heat. This cross-climate perspective is essential for understanding which regions face the greatest climate risks and for designing appropriately tailored adaptation policies.

Third, our approach represents an important step toward linking micro-level evidence with macro-level climate-economy relationships. Aggregate studies relating GDP growth to temperature at the country level ([Dell et al., 2012](#); [Burke et al., 2015](#); [Bilal and Känzig, 2024](#)) have yielded valuable insights but cannot identify the specific mechanisms driving these relationships or the heterogeneity in impacts across different types of firms and workers. Our firm-level analysis reveals the distributional consequences of weather extremes - which types of firms are most vulnerable, which sectors face the greatest disruptions, and how firm characteristics such as size and productivity shape resilience to climate shocks. These microeconomic patterns are crucial for understanding how weather extremes might affect aggregate productivity growth, market structure, and inequality. By documenting the channels through which weather extremes affect firm performance and the substantial heterogeneity in these effects, our findings provide a foundation for more realistic modeling of climate impacts in integrated assessment models and for more targeted policy responses that account for differential vulnerabilities across the economy.

Fourth, in a supplementary analysis reported in the Appendix, we directly test for firm-level adaptation by decomposing weather variables into transitory shocks and long-run climate conditions. We find evidence of partial adaptation to temperature variability and precipitation, but no adaptation to extreme

heat: firms in historically hot locations remain equally vulnerable to heatwave shocks, pointing to hard limits in heat adaptation at the firm level.

The paper is structured as follows. Section 2 introduces the firm data and the weather extremes data. Section 3 describes the empirical strategy and Section 4 reports the results on the effects. Section 5 concludes.

2 Data

Firm data Our primary data source is the Orbis Global Database (Bureau van Dijk, October 2024 edition), which provides comprehensive firm-level financial and operational information for millions of companies worldwide. We extract two main data components: firmographics containing time-invariant company characteristics (legal form, industry classification, addresses, and contact information) and financials history containing balance sheet items, income statement variables, and employment data. Following KalemlÄř-ÄŰzcan et al. (2024), we implement quality control filters to construct a nationally representative firm-level panel. We restrict our sample to yearly financial reports (12-month accounting periods) covering 2000-2022 and exclude firms that ever report negative values for total assets, employment, sales, or tangible fixed assets, as these likely reflect data quality issues rather than economic realities. For firms with multiple financial statements per year (due to different consolidation codes or filing types), we retain the most informative observation using a priority system that favors unconsolidated statements (U2, U1) over consolidated ones (C1, C2) and annual reports over local registry filings, with tie-breaking based on fiscal year-end proximity to December 31st.

To match firm locations with climate data, we implement a three-stage geocoding procedure. First, we geocode full street addresses using the Nominatim geocoding engine with OpenStreetMap data, processing firms country-by-country to manage computational requirements. For addresses where street-level geocoding fails, we implement a second round using postal code and city information to obtain less precise but still useful coordinates. Finally, for remaining firms, we convert the latitude and longitude coordinates provided in the original Orbis database from degrees-minutes-seconds format to decimal degrees. We prioritize coordinate sources in descending order (street-level, postal code-level, Orbis-provided) and merge this spatial information with our firmographics data. Only firms successfully geolocated through at least one of these three methods are retained for analysis, ensuring all observations can be matched to spatially-explicit climate variables.

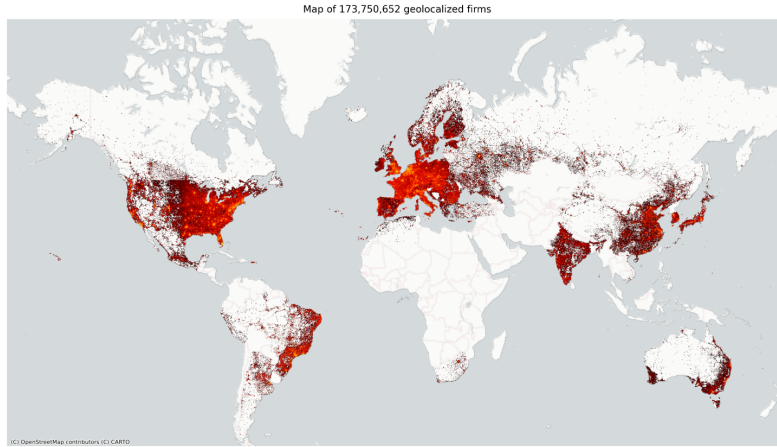


Figure 1: Unique firm locations geocoded from ORBIS.

Our data cleaning and harmonization procedures address several key challenges. Industry classifications from multiple taxonomies (NACE, NAICS, US SIC) are consolidated by extracting two-digit codes and selecting the modal classification when conflicts arise, prioritizing core over primary and secondary designations. At the observation level, we remove firm-years with missing values across four key indicators (total assets, operating revenue, sales, and employment) and apply outlier filters to our dependent variable to eliminate measurement error. These restrictions are intended to exclude uninformative or poorly measured observations that do not contribute meaningfully to identifying within-firm variation. We also exclude firms with fewer than 5 years of data and with more than 250 employees. These criteria are designed to remove very small firms, for which accounting data are often noisy and labor productivity cannot be measured reliably, as well as very large firms that are more likely to operate multiple establishments and complex subsidiary structures. In the latter case, both the assignment of local weather conditions to firm outcomes becomes imprecise and consolidated financial statements may mask establishment-level performance, reducing the reliability of firm-level measures.

Our main outcome variable is labor productivity, which, in its broadest interpretation, captures the efficiency with which inputs are transformed into outputs. Our preferred measure of labor productivity is turnover per employee, as it is widely available across countries and firm sizes in the ORBIS database. For robustness, we also consider alternative specifications including value added per employee and imputed value added per employee (calculated as the sum of labor costs and EBITDA), as proposed by [Gal \(2013\)](#). Control variables include firm size categories (five bins from 5 to 250 employees), age categories (young: <5 years, mature: 5-10 years, old: 10+ years), profitability (EBITDA to total assets ratio),

and capital intensity (fixed assets to total assets ratio), all lagged by one period to mitigate simultaneity concerns. We merge these firm-level data with gridded climate variables (temperature extremes, precipitation patterns, drought indices) matched to firm coordinates by year, creating a final panel dataset that links annual financial performance to location-specific climate realizations. This integrated dataset enables estimation of firm-level climate impacts while controlling for time-invariant firm characteristics, as well as time-varying operational conditions through our constructed control variables.

In total, our dataset covers more than 50 countries (Table 6), though coverage varies considerably across them. Most observations come from European Union countries, supplemented by non-EU European nations such as Great Britain, Serbia, and North Macedonia. Outside Europe, we have a substantial number of observations from South Korea, along with smaller contributions from Japan, India, and Australia. In terms of temporal coverage, our data span the years 2000 to 2022. Orbis provides a consistent amount of firm-level data only from the 2000s onwards, and the number of observations per year increases steadily throughout this period. In the most recent years, the number of observations drops slightly, likely reflecting that some data are still being consolidated. Across the entire timeframe, however, we always maintain at least 300,000 observations per year, ensuring robust coverage for our analysis. A full country coverage breakdown, as well as a year by year breakdown are provided in the appendix (Table 6 and Figure 2 respectively).

Weather extremes We construct annual climate indicators from daily data from the European Centre for Medium-Range Weather Forecasts Reanalysis 5 (ERA-5). We combine hourly data for mean, maximum, and minimum 2-meter temperature and total precipitation, aggregated to daily values on a $0.25^\circ \times 0.25^\circ$ grid. The indicators include: mean annual temperature (TM) and total precipitation (RR); heatwave severity (HW), calculated as the sum of positive temperature anomalies ($T_x - T_{x90th}$) for the whole duration of the most severe heatwave event of the year, where events are defined as periods of at least five consecutive days with maximum temperature exceeding both the 90th percentile (relative to 1981-2010) and 30°C absolute threshold - this calculation is performed over the six warmest months (April-September in the Northern Hemisphere, October-March in the Southern Hemisphere); cold waves severity (CW), calculated using the same logic as for heatwaves, but for periods of at least 5 days with temperature below 0°C and below the 10th percentile; wet days (WD), when daily rainfall exceeds 1mm; and daily temperature variability (TVAR), computed as the root mean squared difference between

daily mean temperature and the corresponding long-term daily average. The full description of the variables is in Table 4.

Table 1: Descriptive Statistics

	Min	P ₅	P ₂₅	Median	Mean	P ₇₅	P ₉₅	Max	SD
Mean Temperature	-16.0	3.7	8.6	12.2	11.6	15.1	17.7	29.4	4.5
Temperature Variability	0.55	1.83	2.34	2.79	2.91	3.41	4.43	7.30	0.79
Heatwaves	0.0	0.0	0.0	0.0	12.1	17.3	56.7	266.5	23.7
Coldwaves	0.0	0.0	0.0	0.0	11.4	13.3	62.0	285.0	23.2
Extreme Rainfall	1.03	36.17	49.76	67.08	82.01	98.11	182.59	765.08	48.28
Extreme Snowfall	0.00	0.00	3.13	11.78	13.40	19.83	35.57	234.93	12.56
Number of Employees	5	5	8	15	29	33	109	249	37

Note: Full description of weather variables is in Table 4.

3 Empirical strategy

We follow and expand on Costa et al. (2024), who use firm-level data from the ORBIS database to estimate the effects of heatwaves on firm productivity. The estimating equation is:

$$\ln(y_{i(gsc)t}) = \mathbf{W}'_{i(gsc)t}\beta + \mathbf{X}'_{i(gsc)t-1}\gamma + \delta_i + \zeta_{sct} + \varepsilon_{i(gsc)t} \quad (1)$$

where i, g, s, c, t index firm, location, sector, country, and year, respectively. The dependent variable $\ln(y_{i(gsc)t})$ is log of the productivity per employee. $\mathbf{W}'_{i(gsc)t}$ is a vector of annualized weather conditions: mean temperature and its square, temperature variability, heatwaves, coldwaves, extreme rainfall, extreme snowfall. We follow Costa et al. (2024) and control for lagged firm characteristics: the number of employees; firm age; profitability, defined as EBITDA over total assets; and capital intensity, defined as fixed assets over total assets. Granular fixed effects are assumed to absorb the remaining unobserved sources of bias. Firm-level fixed effects δ_i reduce the identifying variation to within-firm variation over time. Sector-country-year fixed effects ζ_{sct} further absorb year-specific shocks that are specific to a given sector in a given country. $\varepsilon_{i(gsc)t}$ is the error term. We cluster standard errors at the level of variation of the variables of interest, that is the grid cell level.

To investigate whether firms are able to adapt to recurrent climate hazards, we amend equation (1) by creating a transitory shock component and a long-run climate component. Specifically, for each weather variable we define the shock as the deviation of the annual realization from its long-run mean at the grid-cell level, while the climate variable corresponds to this long-run mean. We then interact each weather shock with its corresponding climate measure. The intuition, in line with [Traeger \(2025\)](#)'s work, is that firms operating in locations that are persistently exposed to a given climatic condition may have adapted to these climatic conditions, but they might still struggle to adapt to unexpected deviations from these conditions. In this framework, the coefficient on the weather shock captures the average effect of unexpected deviations, while the interaction term provides evidence of adaptation if it significantly offsets the shock effect. The specification maintains firm fixed effects and country-sector-year fixed effects as in the baseline model so as to exploit within firm variation over time and is expressed in equation 2.

$$\ln(y_{i(gsc)t}) = \mathbf{W}_{i(gsc)t}^{\text{shock}'} \beta + \left(\mathbf{W}_{i(gsc)t}^{\text{shock}} \times \mathbf{W}_g^{\text{clim}} \right)' \alpha + \mathbf{X}_{i(gsc)t-1}' \gamma + \delta_i + \zeta_{sct} + \varepsilon_{i(gsc)t}, \quad (2)$$

4 Results

Following [Dell et al. \(2012\)](#), we exploit within-unit weather variation to identify causal effects while controlling for time-invariant confounders and common shocks. Estimates from Eq. 1 are presented in Table ???. The dependent variable is the log of turnover per employee. All weather variables are standardized, so coefficients represent the percentage change in firm productivity associated with a one-standard-deviation increase in each weather dimension. The sample comprises 20,546,849 firm-year observations.

The relationship between temperature and firm productivity exhibits the inverted U-shape documented in the macroeconomic literature. The coefficient on mean annual temperature is positive and significant ($\beta = 0.023, p < 0.05$), while the quadratic term is negative and marginally significant ($\beta = -0.014, p < 0.10$). Together, these coefficients imply a concave relationship between temperature and productivity, consistent with country-level estimates from [Burke et al. \(2015\)](#) and regional estimates from [Kalkuhl and Wenz \(2020\)](#). At the firm level, [Zhang et al. \(2018\)](#) document a similar inverted U-shape for total factor productivity across half a million Chinese manufacturing plants, while [Somanathan et al. \(2021\)](#) show that high temperatures reduce both worker productivity and attendance in Indian manufacturing. Notably, the significant linear term in our specification suggests that, at the sample mean,

warmer years are still associated with higher productivity, with the relationship turning negative only at higher temperatures.

Day-to-day temperature variability shows a negative but statistically insignificant association with firm productivity ($\beta = -0.0023$, s.e. = 0.0017). While the sign is consistent with the macro-level findings of [Kotz et al. \(2021\)](#) and [Linsenmeier \(2023\)](#), the lack of statistical significance in our full sample suggests that the effect of temperature variability may be more localized. As we show in the climate zone analysis below, this aggregate null result masks significant heterogeneity across climatic regions.

Turning to extreme events, heatwave severity emerges as the most economically significant weather determinant of firm productivity ($\beta = -0.00246$, $p < 0.01$), implying that a one-standard-deviation increase reduces turnover per employee by approximately 0.25%. To provide an intuitive interpretation, consider a location where the 90th percentile of daily maximum temperature (TX90th) is 30°C, and assume the most severe heatwave lasts 5 days. A one-standard-deviation increase implies that, on average, daily maximum temperatures during the heatwave are 4°C above the TX90th percentile, reaching 34°C. Hence, a 5-day heatwave with an average daily maximum of 34°C would result in a 0.25% reduction in annual turnover per employee.

Our estimated impact is broadly consistent with [Costa et al. \(2024\)](#), who use threshold-based measures across 23 advanced economies and find that ten additional days above 35°C reduce productivity by 0.3% and that one additional 7-day heatwave causes a 0.2% reduction. Despite the different operationalizations, both approaches confirm that heat extremes significantly harm firm-level productivity.

Coldwaves also have a significant negative effect on firm productivity ($\beta = -0.00193$, $p < 0.01$), implying a 0.19% reduction in turnover per employee per standard-deviation increase. This is a notable departure from earlier specifications where cold spells showed a marginally positive association, and it highlights the economic cost of sustained cold extremes on firm operations.

Neither extreme rainfall ($\beta = -0.000234$, n.s.) nor extreme snowfall ($\beta = 0.000828$, n.s.) shows a statistically significant association with productivity in the full sample.

We next examine whether weather impacts vary across climatic regions by estimating Eq. 1 separately for firms located in temperate, cold, and arid Köppen climate zones. Results are reported in Table 2.

The most striking finding is that heatwaves are harmful across all three climate zones: the coefficient is -0.0023 ($p < 0.01$) in temperate zones, -0.0019 ($p < 0.05$) in cold zones, and -0.0017 ($p < 0.05$) in arid zones. This universality underscores the pervasive economic threat of extreme heat regardless of baseline climate.

Temperature variability, which is insignificant in the pooled sample, reveals a strongly significant negative effect in temperate zones ($\beta = -0.0062$, $p < 0.01$), while it is insignificant in cold and arid climates. This suggests that firms in temperate regions, where weather is less extreme on average, may be less equipped to cope with unexpected temperature fluctuations, while firms in more extreme climates have already internalized higher baseline variability.

Coldwaves exhibit a clear geographic pattern: they are strongly negative and significant in cold zones ($\beta = -0.0032$, $p < 0.01$) and arid zones ($\beta = -0.0029$, $p < 0.01$), but insignificant in temperate regions. This is consistent with the interpretation that sustained cold extremes are most disruptive in climates where they are severe enough to cause operational interruptions. In cold zones, the positive and significant quadratic temperature term ($\beta = 0.0143$, $p < 0.05$) suggests a U-shaped local relationship, where both warm and cold deviations from the optimum reduce productivity. Finally, extreme snowfall shows a significant negative effect only in temperate zones ($\beta = -0.00085$, $p < 0.05$), where snowfall events are less frequent and firms may be less prepared.

We report supplementary adaptation results in the Appendix.

Table 2: Estimates of the effects of weather on firm performance.
Coefficients are standardized.

	(1)
	Log(turnover per employee)
Mean Temperature	0.023 06** (0.009 38)
(Mean Temperature) ²	−0.013 53* (0.007 59)
Temperature Variability	−0.002 26 (0.001 71)
Heatwaves	−0.002 455*** (0.000 456)
Coldwaves	−0.001 93*** (0.000 52)
Extreme Rainfall	−0.000 234 (0.000 460)
Extreme Snowfall	0.000 828 (0.000 548)
Num.Obs.	20 546 849
R ₂	0.912
R ₂ Adj.	0.902
RMSE	0.40
Financial Controls	Yes
Firm Fixed Effects	Yes
Country-Sector-Year Fixed Effects	Yes

* p < 0.1, ** p < 0.05, *** p < 0.01. Standard errors clustered by location in parentheses.

Table 3: Subsetting By Climate Zone

	Temperate	Cold	Arid
Mean Temperature	0.002 28 (0.014 93)	0.009 20 (0.007 37)	−0.0045 (0.0188)
(Mean Temperature)^2	0.004 63 (0.015 59)	0.014 30** (0.006 85)	−0.006 06 (0.017 22)
Temperature Variability	−0.006 15*** (0.001 49)	0.003 22 (0.002 10)	0.004 04 (0.003 57)
Heatwaves	−0.002 302*** (0.000 433)	−0.001 857** (0.000 836)	−0.001 742** (0.000 787)
Coldwaves	−0.000 147 (0.000 362)	−0.003 159*** (0.000 898)	−0.002 933*** (0.000 716)
Extreme Rainfall	−0.000 609 (0.000 394)	0.000 552 (0.001 506)	0.000 341 (0.000 469)
Extreme Snowfall	−0.000 853** (0.000 423)	0.001 67 (0.001 13)	0.001 04 (0.001 55)
Num.Obs.	11 440 829	7 693 660	1 393 133
R ₂	0.896	0.888	0.829
R ₂ Adj.	0.885	0.874	0.812
RMSE	0.31	0.52	0.32

* $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$. Controls for firm size (employees) and age class dummies included but not shown.

5 Conclusion

Understanding the impact of climate change on firms is key for assessing the true economic costs of global warming. While recent macroeconometric estimates have shown potentially very disruptive GDP loss estimates, using micro-data is important to understand and validate potential macroeconomic shocks with microeconomic impacts. In this study, we build a global micro panel of over 20 million firm-year observations across 38 countries.

Four main findings emerge. First, we confirm the inverted U-shaped relationship between temperature and turnover per employee documented in both the macro- and microeconomic literature, with a significant positive linear term and a negative quadratic term. Second, we identify heatwaves and coldwaves as the dominant weather determinants of firm productivity: a one-standard-deviation increase in heatwave severity reduces turnover per employee by 0.25%, while coldwaves reduce it by 0.19%. These findings reinforce recent evidence from [Costa et al. \(2024\)](#) on the economic cost of extreme heat and reveal that cold extremes—previously overlooked—also impose significant costs. Third, decomposing the analysis by Köppen climate zones reveals important heterogeneity: heatwaves are harmful across all zones, temperature variability is significant only in temperate climates, and coldwaves are concentrated in cold and arid regions. Fourth, supplementary adaptation analysis reported in the Appendix suggests partial adaptation to temperature variability and precipitation, but no adaptation to extreme heat, pointing to hard limits in autonomous heat adaptation at the firm level.

These results have important implications for climate policy and damage assessment. The dominant role of heatwaves and coldwaves, combined with the heterogeneity revealed by the climate zone analysis, suggests that integrated assessment models focusing exclusively on mean temperature changes may underestimate the economic costs of climate change. Climate projections indicate that extreme events will increase in both intensity and frequency, implying compounding effects on firm productivity. Our estimates can inform the calibration of damage functions in these models by providing microeconomic foundations for aggregate climate-economy relationships.

Several limitations warrant discussion. Our sample covers predominantly European firms in advanced economies, which may exhibit different climate sensitivities than firms in tropical or developing regions where baseline temperatures are higher and adaptive capacity is lower. The productivity effects we estimate capture impacts on turnover per employee but do not account for potential effects on capital pro-

ductivity, firm entry and exit, or broader general equilibrium adjustments. Additionally, our estimates reflect historical climate variation and may not fully capture the effects of unprecedented temperature extremes outside the observed distribution.

Future research could extend this analysis in several directions. Examining heterogeneity across sectors and firm sizes would shed light on which firms are most vulnerable to climate impacts and which adaptation strategies are most effective. Linking weather shocks to specific channels—such as worker health, absenteeism, or supply chain disruptions—would clarify the mechanisms underlying our reduced-form estimates. Finally, incorporating forward-looking climate projections could help quantify the expected productivity losses under different emissions scenarios and inform firm-level adaptation decisions.

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A Appendix

A.1 Adaptation Analysis

We next turn to the adaptation specification described in Eq. 2. Results are reported in Table ???. Most of the coefficients of the weather shocks mirror the baseline findings, confirming the inverted U-shape relationship between temperature and turnover per employee and that unexpected increases in temperature variability, rainfall, heatwaves, and drought conditions significantly reduce firm productivity.

The interaction terms provide evidence of heterogeneous responses consistent with partial adaptation to certain climate features. For temperature variability, the positive and highly significant interaction ($\alpha = 0.004, p < 0.01$) indicates that firms located in areas characterized by persistently high day-to-day temperature fluctuations experience attenuated productivity losses from short-run variability shocks. A similar pattern emerges for rainfall, where the positive interaction term ($\alpha = 0.002, p < 0.01$) suggests that firms operating in wetter climates are better equipped to cope with year-to-year precipitation deviations. We also find evidence of adaptation to snowfall shocks, with a positive and significant interaction implying that firms in snow-prone regions experience smaller productivity losses from unexpected snowfall events.

In contrast, we find no evidence of adaptation to extreme heat. While heatwave shocks have a strongly negative direct effect on productivity, the interaction between heatwave shocks and long-run heatwave exposure is small in magnitude and statistically insignificant ($\alpha = 0.0004, p > 0.10$). This null result is particularly striking: even firms located in historically heat-prone regions—which have had ample opportunity to invest in adaptive measures such as climate control or adjusted work schedules—remain equally vulnerable to extreme heat events as firms in cooler climates. This finding is consistent with physiological constraints on human labor productivity under heat stress that cannot be easily overcome through technological or organizational adaptation (Raymond et al., 2020). It also aligns with the broad-based evidence from Burke et al. (2024) showing limited systematic adaptation across diverse outcomes, and with Callahan and Mankin (2022) documenting that historical exposure to extreme heat has not prevented substantial and unequal economic damages globally.

Table 4: Variable Definitions

Variable	Definition
<i>Temperature measures</i>	
TN	Minimum temperature, annual mean (degrees)
TM	Mean temperature, annual mean (degrees)
TX	Maximum temperature, annual mean (degrees)
<i>Precipitation and evapotranspiration</i>	
RR	Precipitation, annual sum (mm)
PET	Potential evapotranspiration, annual sum (mm)
<i>Extreme events</i>	
HW	Heatwaves: at least 5 consecutive days with $TX > 90$ th percentile and $TX > 30^{\circ}\text{C}$. Severity for each event summed over days as $TX - TX_{90th}$. Value is LARG, the severity of the biggest event in the year (degrees)
CW	Coldwaves: at least 5 consecutive days with $TN < 10$ th percentile and $TN < 0^{\circ}\text{C}$ (degrees)
SNX	Cumulative extreme snowfall: largest 3-day cumulated value for each year (mm)
TVAR	Daily temperature variability: sum of differences (squared, then rooted) between daily mean temperature and the long-term average for that day (degrees)
WD	Wet days: number of days with precipitation > 1 mm (days)

Table 5: Estimates of the effects of weather shocks on firm performance.

	(1)
	Log(turnover per employee)
Mean Temperature Shock	0.0072** (0.0035)
Mean Temperature ² Shock	−0.0079** (0.0039)
Temperature Variability Shock	−0.0108*** (0.0011)
Rainfall Shock	−0.0019** (0.0009)
Wet Days Shock	0.0011 (0.0010)
Heatwaves Shock	−0.0032*** (0.0003)
Coldwaves Shock	0.0001 (0.0004)
Snow shock	−0.0005 (0.0006)
Drought Shock	−0.0007* (0.0004)
Old Firms	0.0183*** (0.0007)
TM shock $\times$ TM climate	0.0044 (0.0080)
TM ² shock $\times$ TM ² climate	0.0002* (0.0001)
TVAR shock $\times$ TVAR climate	0.0040*** (0.0008)

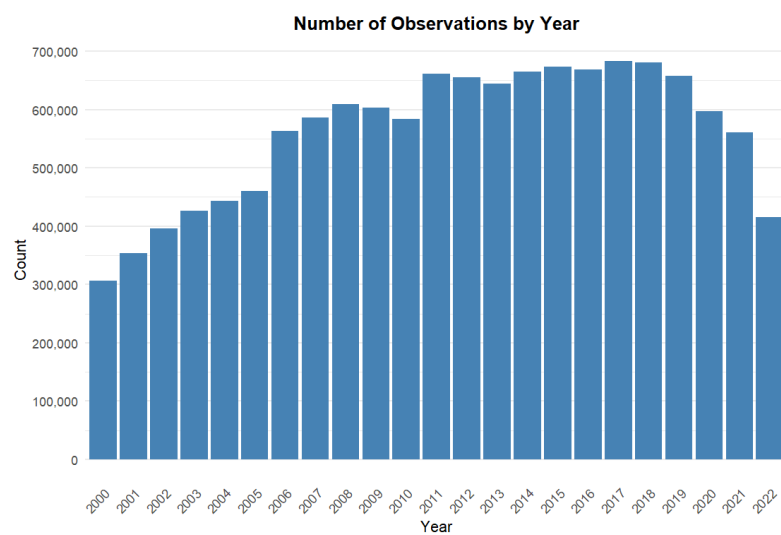


Figure 2: Number of Observations per Year

Table 6: Sample composition by country

Country	N
ES	3,506,349
RU	3,156,216
IT	2,891,460
FR	1,420,176
JP	1,346,199
UA	1,106,633
PT	942,626
SE	696,247
RO	639,445
HU	538,311
BG	505,719
KR	429,858
GB	386,099
HR	320,337
RS	297,392
PL	295,057
FI	283,657
NO	258,836
BE	257,008
GR	237,566
CN	196,775
CZ	186,772
LV	131,510
EE	123,904
SK	112,595
MK	73,457
LT	61,194
DE	58,243
DK	40,495
NL	13,847
ME	13,473
IE	10,760
IS	7,380
AT	2,073
CH	1,272
LU	1,195
AU	849
MT	296
MX	204
TR	62
BR	43
VN	8